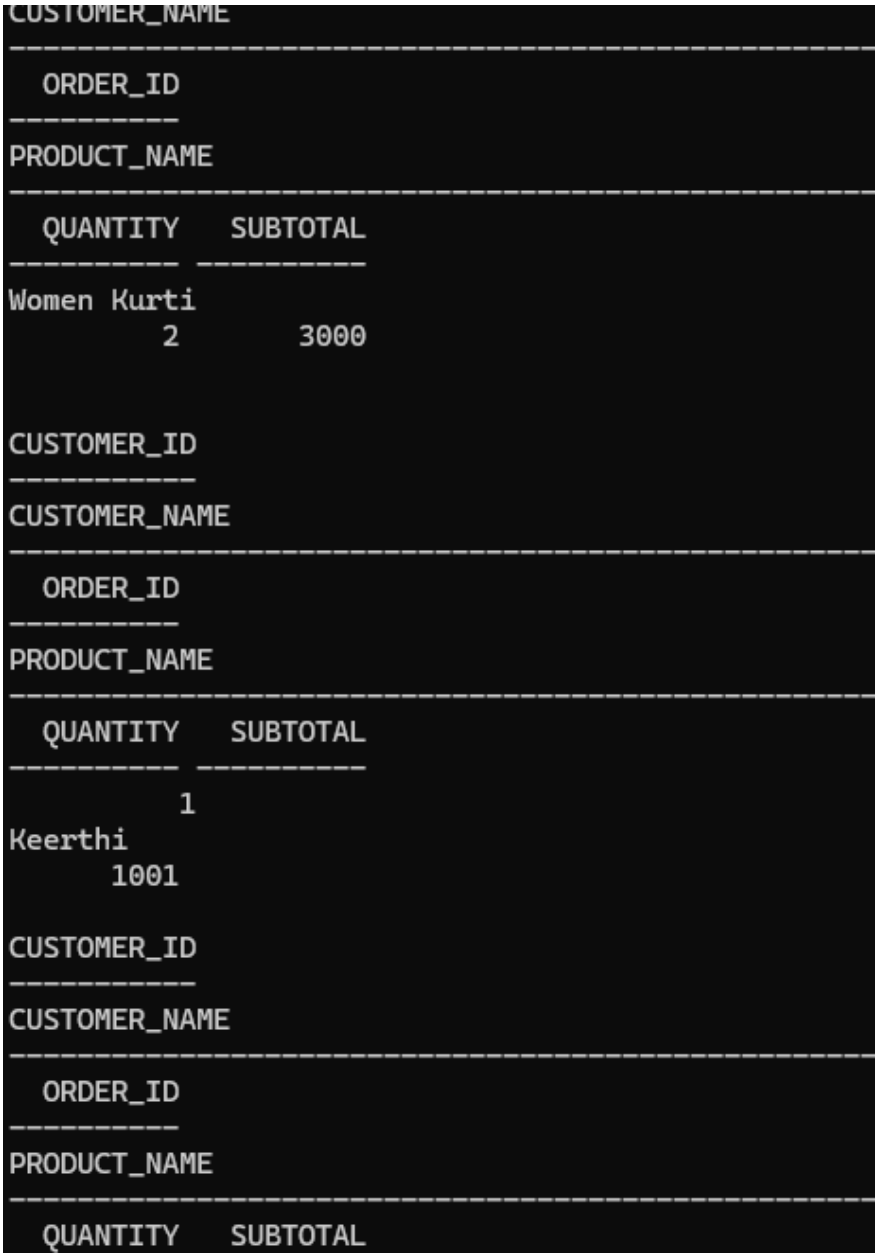
DATABASE RELATIONSHIP ANALYSIS USING JOINS

1. Combining Multiple Tables

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         P.Product_Name,
  5         D.Quantity,
  6         D.Subtotal
  7  FROM Customer C
  8  JOIN Orders O
  9  ON C.Customer_ID = O.Customer_ID
 10  JOIN Order_Details D
 11  ON O.Order_ID = D.Order_ID
 12  JOIN Product P
 13  ON D.Product_ID = P.Product_ID
 14  ORDER BY C.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         1
Keerthi
      1001

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
  ORDER_ID
```



2. INNER JOIN

INNER JOIN displays only matching records from both tables.

2.1 Customer and Orders

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         C.Email,
  4         O.Order_ID
  5  FROM Customer C
  6  INNER JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  ORDER BY C.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------
  ORDER_ID
----------
         1
Keerthi
keerthi@gmail.com
      1001


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------
  ORDER_ID
----------
         2
Priya
priya@gmail.com
```

2.2 Orders and Payment

```
SQL> SELECT O.Order_ID,
  2         P.Payment_ID,
  3         P.Payment_Mode,
  4         P.Payment_Date,
  5         P.Payment_Amount,
  6         P.Payment_Status
  7  FROM Orders O
  8  INNER JOIN Payment P
  9  ON O.Order_ID = P.Order_ID
 10  ORDER BY O.Order_ID;

  ORDER_ID PAYMENT_ID PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT
---------- ---------- -------------------- --------- --------------
PAYMENT_STATUS
--------------------
      1001        501 UPI                  01-OCT-26           2298
Successful

      1002        502 Card                 02-OCT-26            999
Successful

      1003        503 Cash                 03-OCT-26           2499
Successful
```

2.3 Product and Order_Details

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         D.Order_ID,
  4         D.Quantity,
  5         D.Unit_Price,
  6         D.Subtotal
  7  FROM Product P
  8  INNER JOIN Order_Details D
  9  ON P.Product_ID = D.Product_ID
 10  ORDER BY P.Product_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       101
Women Kurti
      1001          2       1500       3000

       102
Men Shirt
      1001          1       2000       2000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------

       103
```

3. LEFT JOIN

LEFT JOIN displays all records from the left table and matching records from the right table.

3.1 Display All Customers and Their Orders

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         C.Email,
  4         O.Order_ID
  5  FROM Customer C
  6  LEFT JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  ORDER BY C.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------
EMAIL
--------------------------------------------------------
  ORDER_ID
----------
         1
Keerthi
keerthi@gmail.com
      1001


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------
EMAIL
--------------------------------------------------------
  ORDER_ID
----------
         2
Priya
priya@gmail.com
```

4. RIGHT JOIN

RIGHT JOIN displays all records from the right table and matching records from the left table.

4.1 Display All Orders and Customer Details

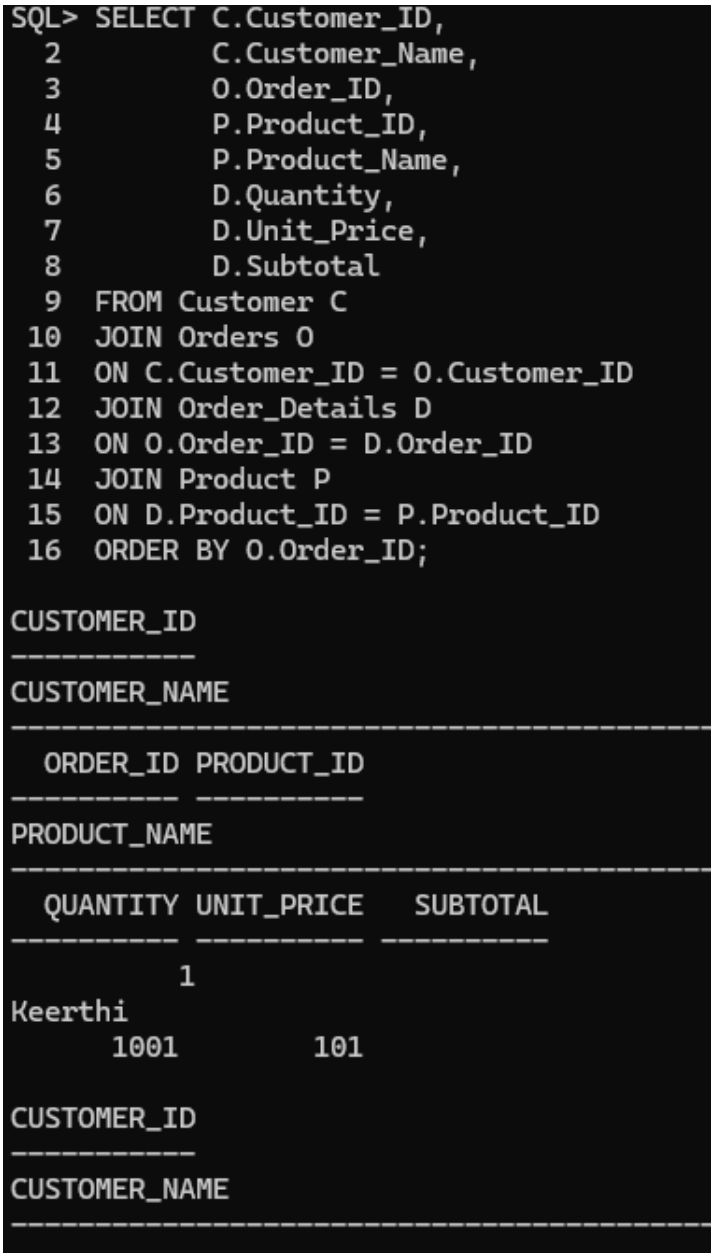
```
SQL> SELECT O.Order_ID,
  2         O.Customer_ID,
  3         C.Customer_Name,
  4         C.Email,
  5         C.Phone
  6  FROM Customer C
  7  RIGHT JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  ORDER BY O.Order_ID;

  ORDER_ID CUSTOMER_ID
---------- -----------
CUSTOMER_NAME
----------------------------------------
EMAIL
----------------------------------------
PHONE
---------------
      1001           1
Keerthi
keerthi@gmail.com
9876543210


  ORDER_ID CUSTOMER_ID
---------- -----------
CUSTOMER_NAME
----------------------------------------
EMAIL
----------------------------------------
PHONE
---------------
      1002           2
Priya
```

5. Complete Order Details

Display customer, order, product and purchase details.



5. Complete Order Details

Display customer, order, product and purchase details.

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         P.Product_Name,
  5         D.Quantity,
  6         D.Unit_Price,
  7         D.Subtotal
  8  FROM Customer C
  9  JOIN Orders O
 10  ON C.Customer_ID = O.Customer_ID
 11  JOIN Order_Details D
 12  ON O.Order_ID = D.Order_ID
 13  JOIN Product P
 14  ON D.Product_ID = P.Product_ID
 15  ORDER BY C.Customer_ID, O.Order_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ----------
         1
Keerthi
      1001

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------\
```

6. Customer Purchase History

Display each customer's purchased products and order details.

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         P.Product_Name,
  5         D.Subtotal,
  6         PY.Payment_Mode,
  7         PY.Payment_Amount,
  8         PY.Payment_Status
  9  FROM Customer C
 10  JOIN Orders O
 11  ON C.Customer_ID = O.Customer_ID
 12  JOIN Order_Details D
 13  ON O.Order_ID = D.Order_ID
 14  JOIN Product P
 15  ON D.Product_ID = P.Product_ID
 16  JOIN Payment PY
 17  ON O.Order_ID = PY.Order_ID
 18  ORDER BY C.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
  SUBTOTAL PAYMENT_MODE         PAYMENT_AMOUNT PAYMENT_STATUS
---------- -------------------- -------------- ----------------
           1
Keerthi
      1001

CUSTOMER_ID
```

7. Customer Purchase History with Payment

Display customer purchase details along with payment information.

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         P.Product_Name,
  5         D.Quantity,
  6         D.Subtotal,
  7         PY.Payment_Mode,
  8         PY.Payment_Status
  9  FROM Customer C
 10  JOIN Orders O
 11  ON C.Customer_ID = O.Customer_ID
 12  JOIN Order_Details D
 13  ON O.Order_ID = D.Order_ID
 14  JOIN Product P
 15  ON D.Product_ID = P.Product_ID
 16  JOIN Payment PY
 17  ON O.Order_ID = PY.Order_ID
 18  ORDER BY C.Customer_ID, O.Order_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- -------------------- ------------------
         1
Keerthi
      1001

CUSTOMER_ID
```

```
-----------
CUSTOMER_NAME
--------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STAT
---------- ---------- -------------------- ------------
Women Kurti
         2       3000 UPI                  Successful


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STAT
---------- ---------- -------------------- ------------
          1
Keerthi
      1001

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------
```

8. Multi-Table Business Report

Display customer, order, product, quantity and payment information in one report.

```
SQL> SELECT PY.Payment_ID,
  2         O.Order_ID,
  3         C.Customer_ID,
  4         C.Customer_Name,
  5         PY.Payment_Mode,
  6         PY.Payment_Date,
  7         PY.Payment_Amount,
  8         PY.Payment_Status
  9  FROM Payment PY
 10  JOIN Orders O
 11  ON PY.Order_ID = O.Order_ID
 12  JOIN Customer C
 13  ON O.Customer_ID = C.Customer_ID
 14  ORDER BY PY.Payment_ID;

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT PAYMENT_STATUS
-------------------- --------- -------------- --------------------
       501       1001           1
Keerthi
UPI                  01-OCT-26           2298 Successful

       502       1002           2
Priya
Card                 02-OCT-26            999 Successful

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT PAYMENT_STATUS
```

9. Payment Transaction Report

Display payment transaction details with customer information.

```
SQL> SELECT PY.Payment_ID,
  2         O.Order_ID,
  3         C.Customer_ID,
  4         C.Customer_Name,
  5         PY.Payment_Mode,
  6         PY.Payment_Date,
  7         PY.Payment_Amount,
  8         PY.Payment_Status
  9  FROM Payment PY
 10  JOIN Orders O
 11  ON PY.Order_ID = O.Order_ID
 12  JOIN Customer C
 13  ON O.Customer_ID = C.Customer_ID
 14  ORDER BY PY.Payment_ID;

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------
PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT PAYMENT_STATUS
-------------------- --------- -------------- ----------------
       501       1001           1
Keerthi
UPI                  01-OCT-26           2298 Successful

       502       1002           2
Priya
Card                 02-OCT-26            999 Successful
```